

AI ENGINEER

Nguyen Hoan Thien

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EDUCATION

Vietnam-Korea University of Information and Communication Technology (VKU) 2021 – 2026
Engineer in Global Information Technology GPA: 3.47/4.0

EXPERIENCE

FPT Software

Junior AI Engineer

Da Nang, Viet Nam

Nov. 2025 – June. 2026

Multi-Agent AI System for Functional Testing

- Developed an **end-to-end Multi-Agent AI System** for automated functional testing in the **automotive domain**, orchestrated with **LangChain** and **LangGraph** for autonomous multi-step decision-making.
- Implemented a **RAG** architecture with vector search over technical documentation and specifications, and integrated **OpenAI GPT models** to generate dynamic test cases and structured reports, achieving high automated execution accuracy.
- Built a full-stack solution (**FastAPI** backend + **React** dashboard) with **AWS Textract** for document extraction, plus robust error handling, logging, and monitoring.
- Technologies:** Python, FastAPI, LangChain, LangGraph, RAG, OpenAI API, AWS Textract, React, Docker

Atomi Cowork – Agentic Knowledge-Work Product

- Built an **agentic knowledge-work product** (similar to Claude Cowork) that autonomously generates professional deliverables: **PowerPoint slides**, **Word documents**, **Excel spreadsheets**, and **landing pages** from natural-language requests.
- Designed a **tool-use agent framework** with multi-step planning, context management, and self-correction, orchestrating document-generation tools to complete complex tasks with minimal human intervention.
- Developed and deployed a **Microsoft Teams App/Bot** (Bot Framework) that embeds the agent directly into Teams, enabling in-context content creation and collaboration within users' daily workflows.
- Technologies:** Python, LangGraph, OpenAI API, Tool-Use/Function Calling, python-pptx/docx/openpyxl, Microsoft Teams Bot Framework, FastAPI, Docker

AI Meeting Assistant

- Developed an **AI meeting assistant** that records and transcribes meetings, generates concise **summaries**, and extracts **action items** and key decisions.
- Built an **ASR speech-to-text** pipeline with speaker diarization, and applied **LLM-based summarization** over transcripts for searchable, structured meeting notes.
- Implemented **semantic search** over meeting history using a vector database, enabling natural-language querying of past discussions.
- Technologies:** Python, FastAPI, ASR (Speech-to-Text), OpenAI API, Vector DB, React

Center for Statistical Data Processing and Integration – Da Nang Branch

AI Engineer Intern

Da Nang, Viet Nam

May. 2025 – Oct. 2025

- Built an **end-to-end AI search system** to identify Vietnamese **VSIC/VCPA economic codes** from text descriptions, achieving high retrieval accuracy and optimized search performance.
- Processed and cleaned over **6 million records** (duplicate removal, missing value imputation, text normalization, label encoding) to create a high-quality training dataset.
- Fine-tuned **PhoBERT** with transformer encoder architecture combining text embeddings and frequency features, using **mixed precision**, **gradient accumulation**, and **hyperparameter tuning**, improving model accuracy and training efficiency.
- Optimized and deployed the model via **ONNX Runtime** with **quantization** (INT8/FP16) for mobile compatibility, reducing model size and latency.
- Developed **RESTful APIs** with **FastAPI** for real-time VSIC/VCPA search, data upload, monitoring, and offline model downloads, with robust validation, logging, and error handling.

- Created a search interface and admin dashboard for uploading/validating new data, enabling real-time predictions and offline capability in low-connectivity environments.
- Implemented **ASR speech-to-text** service to allow voice-based input for AI-powered VSIC/VCPA code search.
- Developed a **Flutter mobile application** integrating **onnxruntime** to deploy the quantized PhoBERT model on Android and iOS devices for on-device AI inference.
- Monitored and improved performance metrics (Recall@K, QPS, latency p50/p95), enhancing user experience and search reliability.
- **Technologies:** Python, FastAPI, PhoBERT, ONNX Runtime, Quantization, Flutter, Dart, Docker, Git

SafeHorizons Software Services Co., Ltd

Da Nang, Viet Nam

Python Backend Developer

Aug. 2023 – June 2024

- Designed and developed a full-scale **agricultural marketplace** from scratch, directly connecting farms with consumers and investors, contributing to a successful platform launch and **rapid user adoption**.
- Implemented an efficient **background task processing** system and **caching mechanism**, resulting in a **40%** reduction in page load time and **real-time** product availability updates.
- Identified and resolved over **50 critical bugs** in the platform's help and support system, significantly improving **stability** and **user satisfaction**.
- Optimized **database queries** and **API endpoints**, achieving a **25%** reduction in transaction processing time and enhancing system **reliability** under load.
- Developed comprehensive **automated test coverage**, leading to a **40%** decrease in production incidents and improved **code quality**.
- Leveraged **Docker** and **CI/CD pipelines** for consistent and automated deployments across development and production environments, including **nitrogen printing systems** for agricultural documentation.
- **Technologies:** Django REST Framework, PostgreSQL, Redis, Celery, JWT, Docker, Git, CI/CD

PROJECTS

Emergix - Unified Emergency Operations & Planning Platform

November – December 2024

The 33rd Vietnam IT Student Olympiad - PROCON VIETNAM

Third Prize - Open Source Software Category

- Participated in **brainstorming** and **conceptualizing** ideas for the competition
- Analyzed, designed, and developed the **Communities Module** (one of three core modules)
- Developed the **LLM Service** using **GPT API** with **RAG** and **FAISS** as the Vector Database
- Built the **Notification Service** using **SMTP** (email) and **Twilio** (SMS) for **real-time alerts**
- Implemented **containerization** and deployment using **Docker**
- Wrote comprehensive **documentation** for maintainability and knowledge sharing
- **Technologies:** Appsmith, Strapi, Dify, SMTP, Twilio, Python, JavaScript/TypeScript
- **GitHub:** github.com/vku-open-source

GuardianNet – AI-powered Online Protection Tool

April 2025

GDGOC Hackathon Vietnam 2025

Top 5 in Central Region, Team Leader

- Built a **browser extension** (Chrome, Edge, Brave) integrating a **multi-task AI model** to automatically scan and blur images containing violent content while detecting human presence before display
- Combined **text processing** to filter profanity and abusive language, allowing hiding or replacing with **[**]** characters
- Provided Options interface for users to configure **sensitivity levels**, whitelist/blacklist, and a **transparent dashboard** for parents/teachers to view detailed reports
- Designed **Responsible AI architecture** ensuring fairness, transparency, and user data privacy protection
- Trained a multi-task model (violence detection + human detection) on **MobileNetV2** achieving **98.5% accuracy**
- Developed the extension with **Plasmo** (Manifest V3, content script & background script) for **real-time processing**
- **Technologies:** Multi-task Learning, MobileNetV2, TensorFlow/Keras, Plasmo, JavaScript/TypeScript, Python
- **GitHub:** github.com/Thiennnguyen21it/GDGOC-hackathon-2025-AIGenerated

TECHNICAL SKILLS

Programming language: Python, JavaScript/TypeScript, Dart
AI/ML: LangChain, LangGraph, RAG, OpenAI API, PhoBERT, TensorFlow/Keras, ONNX Runtime
Backend: Django, FastAPI, Flask, NodeJS
Frontend/Mobile: ReactJS, Flutter
Database: PostgreSQL, MySQL, Firebase, MongoDB
Cloud & DevOps: AWS (Textract), Linux, Docker, Git, GitHub, GitLab, CI/CD
Languages: Vietnamese (Native), English (TOEIC 750 - Fluent in English)

HONORS & AWARDS

Google Developers Groups On Campus Hackathon Vietnam 2025 <i>Top 5 in Central Region, Team Leader</i>	May 2025
Award for Outstanding Student Achievement – 2024 <i>Vietnam-Korea University of Information and Communication Technology</i>	December 2024
Third Prize - Open Source Software Category <i>The 33rd Vietnam IT Student Olympiad - PROCON VIETNAM</i>	December 2024
VKU Robocar Innovation Contest <i>Consolation Prize</i>	May 2023
English Talent Competition VKU <i>Third Prize</i>	October 2022